

ADAM ECCLES

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SUMMARY

Full Stack Software Engineer with a background as a former world-class esports athlete, bringing strong problem-solving and leadership into software development. Experienced in building scalable web applications and production AI systems, including LLM-powered agents that reduced student failure rates by 50%+. Proficient in TypeScript, JavaScript, and Python, with hands-on experience in Node.js, Express, PostgreSQL, and Next.js. Skilled in developing secure APIs, integrating machine learning workflows, and delivering data-driven, end-to-end solutions in production environments

WORK EXPERIENCE

Forward Deployed Engineer

May 2026 - Present

- Led discovery for client engagements, translating business needs into technical architecture, process designs, and end-to-end user journeys.
- Drove engineering delivery by defining and assigning user stories to senior engineers and validating implementations against acceptance criteria.
- Automated enterprise-scale workflows that previously required days of manual effort, reducing operational overhead and improving execution consistency.
- Built a batch orchestration system to deploy Python workers with assigned tasks, enabling automated execution of distributed workloads.

Full Stack - Software Engineer - Automate Army

Jan 2026 - May 2026

- Built a TypeScript (strict) + Node.js + Express backend API with layered architecture (Routes/Controllers/Services) for a materials operations dashboard.
- Implemented JWT auth, RBAC (admin middleware), and bcrypt password security.
- Developed data layer with Drizzle ORM + PostgreSQL and migration-based schema management.
- Added Zod request validation, standardized API error handling, and typed response contracts.
- Integrated Google BigQuery analytics (filtering, pagination, CSV export, timeout/error handling).
- Added observability and quality controls with Winston structured logging, Vitest tests, and Docker deployment support.
- Built a modular production reporting form with Next.js, TypeScript, React Hook Form, and Zod for complex nested operational data entry.
- Implemented typed React Query + Axios API integration with structured error handling and field-level validation feedback.
- Added debounced draft auto-save/recovery (localStorage) and reusable responsive sections to improve completion reliability and UX.

AI Agent Builder - Maryville University

Sep 2025 - May 2026

- Built NLP-powered AI agents using retrieval-augmented generation (RAG) to deliver citation-based responses from institutional knowledge bases, reducing hallucinations and improving response accuracy.
- Engineered RESTful integrations with the CampusLabs Engage Events API to automate event data retrieval with 100% response reliability.
- Engineered ETL pipelines to process and analyze 50K+ student transcript records, extracting skills via NLP and mapping them into a Foundational–Advanced skills taxonomy.
- Performed agent analytics on large-scale chat logs, measuring agent load, user volume, and redaction rates, and visualized usage trends with matplotlib to identify peak demand periods.
- Delivered actionable skill insights that help students identify strengths, close learning gaps, and align coursework with career needs.
- Integrated external APIs (CampusLabs Engage Events API) into RESTful services, achieving 100% reliable data ingestion for automated event analytics.

AI Agent Intern - NeuralSeek

Aug 2025 – Sep 2025

- Developed and deployed agentic AI systems leveraging LLMs for real-world customer use cases on AWS using Python, enhancing customer interaction efficiency
- Implemented multi-agent Retrieval-Augmented Generation (RAG) workflows and reduced human intervention in enterprise workflows.
- Collaborated with cross-functional teams to design, test, and optimize production-ready AI agents for education, customer support, and enterprise tasks, leading to more effective and reliable AI solutions
- Validated agent responses at 95% factual accuracy using synthetic benchmarks

AI Research Intern - Maryville University

Jan 2025 – May 2025

- Engineered and deployed AI-driven solutions to enhance university operations and student learning.
- Developed an intelligent tutoring systems used in 12 + graduate courses, reducing Drop/Fail/Withdraw rates in Graduate Nursing from 36% to 16%.
- Supported 13,000+ faculty and student interactions through adaptive educational agents.
- Created a Social Learning Companion for Cybersecurity SOC training that improved retention and reduced practical errors.
- Led continuous model-testing and retraining cycles, improving agent accuracy and robustness across deployments.

Lead Instructor - Valorant Program, Saudi Esports Academy

Jun 2024 – Aug 2024

- Designed and delivered an 8-week esports curriculum for 50+ students.
- Organized a professional-grade tournament, creating a structured competitive environment.
- Mentored diverse students, enhancing teamwork and performance under pressure.

Professional Esports Player - Team Liquid, 100 Thieves, Fnatic, NIP

Jan 2016 – Aug 2024

- Achieved #1 world ranking (2020); Recognized by Forbes as one of the top 3 players globally (2020)
- Secured 9 national championships across Europe & North America.
- Led multinational rosters, developing leadership, communication, strategic problem-solving, high intensive decision-making.

PROJECTS

Formula 1 Race Pace Prediction - Machine Learning Project

- Processed and modeled telemetry from 1,000+ race laps and 60+ qualifying sessions to predict lap-level race pace for 20 F1 drivers in the 2025 Canadian Grand Prix.
- Engineered 12+ numerical and categorical features from raw telemetry and weather forecasts, including sector-time aggregates, delta-to-pole, and normalized temperature/rain inputs.
- Using XGBoost, and feature-importance pruning to improve inference speed while maintaining recall.
- Diagnosed model variance via per-fold driver analysis (e.g., 2.07s MAE outlier due to DNF data), improving model robustness through imputation and filtering.
- Built visualization dashboards to display feature importances and driver-level prediction breakdowns.

Siamese Network for Real-Time Face Verification - Machine Learning Project

- Engineered and trained a 39M-parameter Siamese neural network for one-shot face verification using 4096-dim embeddings and L1 distance.
- Engineered and labeled 600+ contrastive image pairs using webcam input (anchor/positive) and LFW negatives.
- Achieved 92.3% face pair classification accuracy with 90% recall on unseen validation identities.
- Reduced memory usage by ~40% via cache(), prefetch(), and mixed_precision, enabling real-time (<1s) verification on CPU.
- Deployed a live OpenCV-based face verification loop supporting real-world inference.

Phishing Website Detection - Machine Learning Project

- Developed a supervised machine learning pipeline to detect phishing websites using 11K+ labeled samples with 30+ engineered features (URL structure, domain metadata, JavaScript behavior, SSL validity).
- Applied SMOTE for class balancing and PCA for dimensionality reduction; trained and evaluated Random Forest and L1-regularized Logistic Regression models using 10-fold cross-validation.
- Conducted model interpretability via feature importance metrics to identify top predictive signals and minimize overfitting.
- Achieved 98% recall and 97.2% accuracy while cutting feature set by 33% and inference time by 60%.

ModelMatch - Agentic AI Model Router (You.com Hackathon Solo Project)

- Built an AI routing system using FastAPI and the You.com Search API to automatically select the most efficient LLM for each task.
- Integrated benchmark-based scoring (LMSYS, HuggingFace, OpenCompass) to rank models like GPT-4, Claude, Gemini, Mistral, and xAI by accuracy, speed, and cost.
- Achieved up to 70% lower compute costs through dynamic routing modes (Balanced, Quality-First, Cost-First) while maintaining high output quality.

MultiModal AI Agent for VISA Assistance - Agentic AI Project

- Built BorderBuddy, an Agentic AI multi-agent workflow that automates visa guidance by integrating legal, health, and compliance agents with verification loops. Leveraged NLP, official .gov/WHO data retrieval, and SMTP integration to generate and deliver citation-backed travel guides in both email and document formats, creating a seamless end-to-end user experience.

FPS Aim Warmup - Browser-based aim trainer (JavaScript, HTML)

- Developed a browser-based 3D FPS aim trainer using JavaScript, Three.js, and Howler.js, implementing modular game architecture, real-time event handling, and raycasting algorithms for precision hit detection and responsive feedback.
- Engineered scalable UI and state-management logic with localStorage persistence, asynchronous input handling, and WebGL rendering optimization, reducing input latency by ~25% and ensuring smooth 60 FPS performance across major browsers.

EDUCATION

B.Sc. Computer Science (GPA: 4.0)

Maryville University — St. Louis, MO (Expected May 2026)

- Full Ride via Esports Program | Machine Learning | Artificial Intelligence | Data Structures & Algorithms | Discrete Mathematics | Deep Learning | Operating Systems | Virtualisation & Cloud Security | Project Management
- Excellence in Computer Science Award Winner.

BTEC Level 3 National Extended Diploma in Computing – Triple Distinction (DDD) - Burnley College — Lancashire, UK (2019)

- Digital Forensics, Networks & Security, Website Development, and Database Design.

TECHNICAL SKILLS & TOOLS

- Languages: Python, JavaScript, Java, SQL, Node.js, Express.js
- Frameworks: React Native, Expo, Node.js, Express, GraphQL, Docker
- ML Frameworks: PyTorch, TensorFlow, scikit-learn
- Data Tools: pandas, numpy, Polars, DuckDB, Parquet, Matplotlib, PostgreSQL, GraphQL, CI/CD
- Cloud & Dev: AWS (S3, Lambda, Glue, Redshift), Google Cloud, BigQuery, GCP
- Certifications: AWS Cloud Practitioner, AWS Cloud Quest, NeuralSeek Agentic AI Level 1–3, You.com AI Agent Masterclass Level 3